26030 – 2022 Winter Exam

Question 1:

(5 points)

Natural gas, which is used for cooking and heating domestically, contains propane (C_3H_8). When propane is burned in the presence of air, it produces CO_2 and steam according to the following reaction:
$C_3H_8 + O_2 \rightarrow CO_2 + H_2O$
How many moles of CO_2 can be produced, if we burn 24 gram of C_3H_8 with 24 gram of O_2?

(a) 0.3 mole
(b) 0.35 mole
(c) 0.4 mole
(d) 0.45 mole
(e) 0.5 mole

Question 2:

(2 points)

How many unpaired electrons does the Molybdenum (Mo) atom have? How many of these electrons are located in d-orbitals?

(a) 5 unpaired electrons and 4 electrons in d-orbitals
(b) 5 unpaired electrons and 5 electrons in d-orbitals
(c) 6 unpaired electrons and 4 electrons in d-orbitals
(d) 6 unpaired electrons and 5 electrons in d-orbitals
(e) 6 unpaired electrons and 6 electrons in d-orbitals

Question 3:

(5 points)

The following reaction is used for the synthesis of ammonia (NH_3) from nitrogen (N_2) and hydrogen (H_2):

$N_2\ (g) + 3H_2\ (g) \rightarrow 2NH_3\ (g)$

The following values have been measured: ΔH^o for NH_3 (g) is -46.11 kJ/mol and ΔS^o for N_2 (g), H_2 (g) and NH_3 (g) are 191.61, 130.79 and 192.4 (J/(mol.K)), respectively.

At 45ºC, calculate the ΔG^o and K for the reaction using the ΔH^o and ΔS^o values given above.

(a) $\Delta G° = + 28.8$ kJ/mol and $K = 5.97 \cdot 10^4$
(b) $\Delta G° = + 28.8$ kJ/mol and $K = 4.97 \cdot 10^{-4}$
(c) $\Delta G° = - 28.8$ kJ/mol and $K = 4.47 \cdot 10^{-4}$
(d) $\Delta G° = - 28.8$ kJ/mol and $K = 5.47 \cdot 10^4$
(e) $\Delta G° = - 28.8$ kJ/mol and $K = 5.47 \cdot 10^{-4}$

Question 4:

(3 points)

Nitrogen monoxide (NO) can be made through the reduction of dilute nitric acid (HNO_3) by elemental copper:
H^+ (aq) + Cu (s) + NO_3^- (aq) → Cu^{2+} (aq) + NO (g) + H_2O (l)

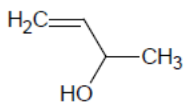
Balance the equation and mark the right set of coefficients for H^+, Cu, NO_3^-, Cu^{2+}, NO and H_2O, respectively:

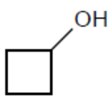
(a) 2, 3, 2, 3, 2 and 1
(b) 8, 2, 3, 2, 3 and 4
(c) 16, 2, 3, 2, 3 and 8
(d) 16, 3, 2, 3, 2 and 8
(e) 8, 3, 2, 3, 2 and 4

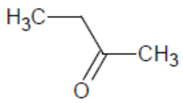
Question 5:

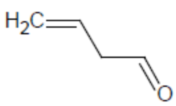
(2 points)

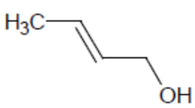
A molecule has the molecular formula C_4H_8O. Which of the structures shown below is *not* consistent with this molecular formula?

(a) 

(b) 

(c) 

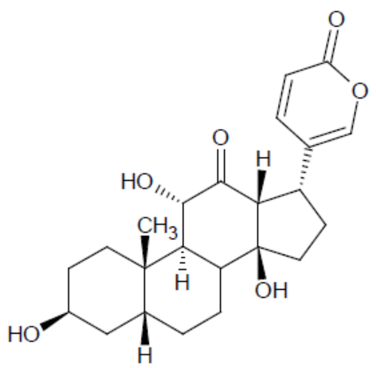
(d) 

(e) 

Question 6:

(5 points)

In the given organic molecule, besides the number of alcohol groups, which other functional groups are present?



(a) Ether, Aldehyde, Alkene
(b) Ester, Ketone, Alkyne
(c) Ester, Ketone, Alkene
(d) Ether, Ketone, Alkene
(e) Ester, Aldehyde, Alkyne

Question 7:

(5 points)

Assuming the complete dissociation of the salt, what is the freezing point of the aqueous 0.75 m solution of $MgCl_2$? Water has a freezing point depression constant of 1.86˚C/m.

(a) 4.18 ˚C
(b) -4.18 ˚C
(c) 3.72 ˚C
(d) -3.72 ˚C
(e) 5.18 ˚C

Question 8:

(5 points)

In which of the following cases are all the 4 salts *not* soluble in water?

(a) LiCl, $Ca(OH)_2$, Na_2CO_3, KNO_3
(b) LiCl, KOH, Na_2CO_3, $Pb(NO_3)_2$
(c) AgCl, KOH, Na_2CO_3, KNO_3
(d) LiCl, $Ca(OH)_2$, $BaCO_3$, KNO_3
(e) AgCl, $Ca(OH)_2$, $BaCO_3$, $SrSO_4$

Question 9:

(5 points)

Which of the following lists of compounds does not contain acids?

(a) CH_3CH_2COOH, NaOH, CH_4
(b) KOH, NaH, H_3PO_4
(c) NH_3, C_2H_5OH, H_2SO_4
(d) NO_2, CH_3CH_2COOH, C_2H_2
(e) C_4H_9OH, NH_3, NaH

Question 10:

(5 points)

A compound with a weight of 108 mg is filled into a balloon and then converted into an ideal gas. The final volume of the gas after conversion is 33.6 mL at 273.15 K and 1.0 atm. What is the molecular formula of the compound?

(a) C_3H_8
(b) C_5H_{12}
(c) C_2H_2
(d) C_4H_{10}
(e) C_2H_6

Question 11:

(5 points)

Ethylene is a sweet smelling organic compound, widely used to force the ripening of fruits, is produced from ethane (C_2H_6). Ethane has a standard boiling point of -88.5 ºC and its molar heat of evaporation is 14.7 kJ/mol. If we want to increase the boiling point to 30 ºC, calculate the minimum pressure needed.

(a) 42 atm
(b) 58 atm
(c) 34 atm
(d) 62 atm
(e) 28 atm

Question 12:

(2 points)

Consider the following reaction:

$2\ ClO_2\ (aq) + 2\ OH^-\ (aq) \rightarrow ClO_3^-\ (aq) + ClO_2^-\ (aq) + H_2O\ (l)$

The following starting concentrations and initial rates of reactions were observed:

Experiment	$[ClO_2]$ (M)	$[OH^-]$ (M)	Initial Rate (M/s)
1	0.030	0.015	0.01240
2	0.010	0.015	0.00138
3	0.010	0.045	0.00414

What is the order of the reaction with respect to OH^- ?

(a) 0
(b) 1
(c) 2
(d) 3
(e) None of the listed.

Question 13:

(3 points)

$FeSO_4$ is reacted with $KMnO_4$ in an acidic aqueous solution. What is $FeSO_4$ converted into?

(a) Fe^{3+} and S
(b) Fe^{2+} and S
(c) Fe^{3+} and SO_4^{2-}
(d) Fe^{3+} and S^{2-}
(e) Fe^{2+} and SO_4^{2-}

Question 14:

(5 points)

Calculate the pH of a 0.030 M ammonia (NH_3) solution. Note that ammonia is a weak base with a $K_b = 1.8 \times 10^{-5}$.

(a) 8.2
(b) 9.9
(c) 11.1
(d) 10.8
(e) 12.1

Question 15:

(5 points)

Calculate the solubility product for AgCl ($K = [Ag^+]\cdot[Cl^-]$) at 25ºC using the standard reduction potentials for the half-cell reactions given below:

(1) $Ag^+ (aq) + e^- \rightarrow Ag (s)$ $E^0 = 0.8$ V
(2) $AgCl (aq) + e^- \rightarrow Ag (s) + Cl^- (aq)$ $E^0 = 0.215$ V

(a) 6.4×10^9 M^2
(b) 1.3×10^{-10} M^2
(c) 7.8×10^9 M^2
(d) 1.9×10^{-10} M^2
(e) 1.6×10^{-10} M^2

Question 16:

(3 points)

For an endothermic equilibrium reaction, $3O_2$ (g) $\rightleftharpoons$ $2O_3$ (g), predict in which direction the reaction will shift (i) if the O_2 is added, (ii) if the temperature is increased and (iii) if the pressure is increased?

(a) (i) right, (ii) right and (iii) left
(b) (i) right, (ii) left and (iii) left
(c) (i) right, (ii) right and (iii) right
(d) (i) left, (ii) left and (iii) right
(e) (i) left, (ii) left and (iii) left

Question 17:

(3 points)

For the sulphate ion, SO_4^{2-} , the sulphur atom is placed centrally surrounded by the four oxygen atoms. Using VSEPR theory determine the geometry of the sulphate ion.

(a) square planar
(b) tetrahedral
(c) octahedral
(d) distorted tetrahedral
(e) distorted octahedral

Question 18:

(2 points)

The boiling point of the NH_3 (-33 ºC) is higher than that of PH_3 (-88 ºC). Which of the following explains this observation?

(a) The dispersion forces are weaker between PH_3 molecules than between NH_3 molecules.
(b) There are weaker dipole-dipole interactions between PH_3 molecules than between NH_3 molecules.
(c) The hydrogen bonds are much stronger between NH_3 molecules.
(d) Both dispersion forces and dipole-dipole interactions are weaker between PH_3 molecules than between NH_3 molecules.
(e) None of the listed.

Question 19:

(5 points)

The following reaction takes place in 1 L of water and has a $\Delta G°$ of -15 kJ/mol at 30 ºC.

$2 A + B \rightleftharpoons 2C + D$

Calculate the moles of D at equilibrium, if A is 0.1 mole, B is 0.1 mole and C is 0.8 mole.

(a) 0.2 mole
(b) 0.3 mole
(c) 0.4 mole
(d) 0.6 mole
(e) 0.8 mole

Question 20:

(2 points)

Consider the following reaction:

$CaBr_2$ (aq) + K_2SO_4 (aq) → $CaSO_4$ (s) + 2KBr (aq)

Which type of reaction does it represent?

(a) Precipitation reaction
(b) Acid base reaction
(c) Redox reaction
(d) Decomposition reaction
(e) None of the listed

Question 21:

(3 points)

Calcium (Ca), potassium (K), argon (Ar), chlorine (Cl) and sulphur (S) can occur as isoelectronic species with the $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ electronic structure.
Which of the following is the correct combination of charges with the isoelectronic species arranged according to a decreased radius.

(a) $S^{2-} > Cl^- > Ar > K^+ > Ca^{2+}$
(b) $Ca^{2-} < K^- < Ar < Cl^+ < S^{2+}$
(c) $Ca^{2+} < K^+ < Ar < Cl^+ < S^{2+}$
(d) $Ca^{2-} > K^- > Ar > Cl^- > S^{2-}$
(e) K > Ca > S > Cl > Ar

Question 22:

(5 points)

In which case are all the mentioned acids weak acids?

(a) H_2SO_4, HCl, HF
(b) HCl, HF, HNO_3
(c) HF, CH_3COOH, HCl
(d) HF, CH_3COOH, HNO_2
(e) CH_3COOH, HNO_2, HNO_3

Question 23:

(5 points)

Yellow Sulphur is burned to produce the SO_2 to be used as preservative in wine making. Calculate the maximum amount of SO_2 (in grams) that can be produced from 12 g of Sulphur in the presence of 25 g of oxygen.

$S\ (s) + O_2\ (g) \rightarrow SO_2\ (g)$

(a) 20.5 g
(b) 18.9 g
(c) 28.6 g
(d) 12.7 g
(e) 23.9 g

Question 24:

(5 points)

The reaction $NO\ (g) + O_3\ (g) \rightarrow NO_2\ (g) + O_2\ (g)$ has a constant activation energy of 63 kJ/mol in the temperature range of 400 – 1000 °C. If the rate constant at 600 °C is k_1 and at 800 °C is k_2, how much faster or slower is the reaction at 800 °C as compared to that at 600 °C, i.e. the ratio of k_2/k_1?

(a) 0.2
(b) 0.5
(c) 5
(d) 50
(e) Not enough information is given.

Question 25:

(5 points)

Upon burning the benzene (C_6H_6) in the presence of the air, carbon dioxide and water are produced. The standard enthalpy of formation of benzene is 49.04 kJ/mol. Calculate the amount of heat released in joules per gram of the benzene burned in air.

The standard enthalpy of formation of CO_2 is -393.5 kJ/mol
The standard enthalpy of formation of water is -285.8 kJ/mol

(a) -39830 J/g
(b) -48410 J/g
(c) -44340 J/g
(d) -41830 J/g
(e) -50320 J/g